CSCI-4041
Spring 2026
Paper Homework 4 - Chapters 6 and 12

Name: Vector Bramwattanan **Section:** _____

1. Consider the max-oriented heap described in chapter 6 of the CLRS textbook.

(a) What are the minimum and maximum numbers of elements in a heap of height h?

$min = 2^h$

$Max = 2^{h+1} - 1$

(b) Show that an n-element heap has height $h = \lfloor \log(n) \rfloor$.

$h = \lfloor \log(n) \rfloor$

$2^h \leq n \leq 2^{h+1} - 1 \rightarrow n \leq 2^{h+1} - 1 < 2^{h+1}$

$\downarrow$ $\qquad\qquad\qquad\qquad \downarrow$

$h \leq \log n \qquad\qquad \log(n) \leq h+1$

$\downarrow$

$h > \log(n) - 1$

$\log(n) - 1 \leq h \leq \log(n)$

(c) Show that in any subtree of a max-heap, the root of the subtree contains the largest value occurring anywhere in that subtree.

* contradiction → root of subtree not the largest

-however by property max heap node key is greater than children key

(root) key > (children 1) key > (children 2) key

* this contradicts with our assumption above

so root of subtree is always the largest

2. Consider the max-oriented heap described in chapter 6 of the CLRS textbook.

(a) Where in a max-heap might the smallest element reside, assuming that all elements are distinct?

- parent is bigger than child

so the smallest element cannot have children

- it is at bottom of tree → leaf

(b) Show that, with the array representation for storing an n-element heap, the leaves are the nodes indexed by $\lfloor n/2 \rfloor + 1, \lfloor n/2 \rfloor + 2, \ldots, n$.

$\lfloor n/2 \rfloor + 1, \lfloor n/2 \rfloor + 2, \ldots,$

leaf → $2i > n$

internal → $2i \leq n$

$2i < n$

$i < n/2$

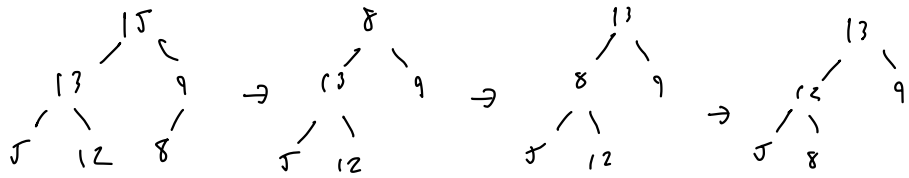
$i \leq n/2$ → $1, 2, \ldots \lfloor n/2 \rfloor$ → not leaf

$2i > n$ → $\lfloor n/2 \rfloor + 1 + \ldots + n$ → leaf

3. Consider the max-oriented heap described in chapter 6 of the CLRS textbook.

(a) Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-EXTRACT-MAX on the following heap.

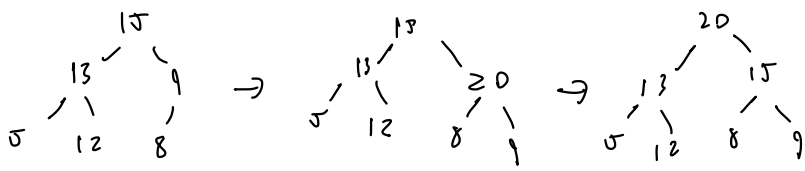
$$A = [15, 13, 9, 5, 12, 8]$$



A = [13, 12, 9, 5, 8]

(b) Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-INSERT(A, 20) on the following heap.

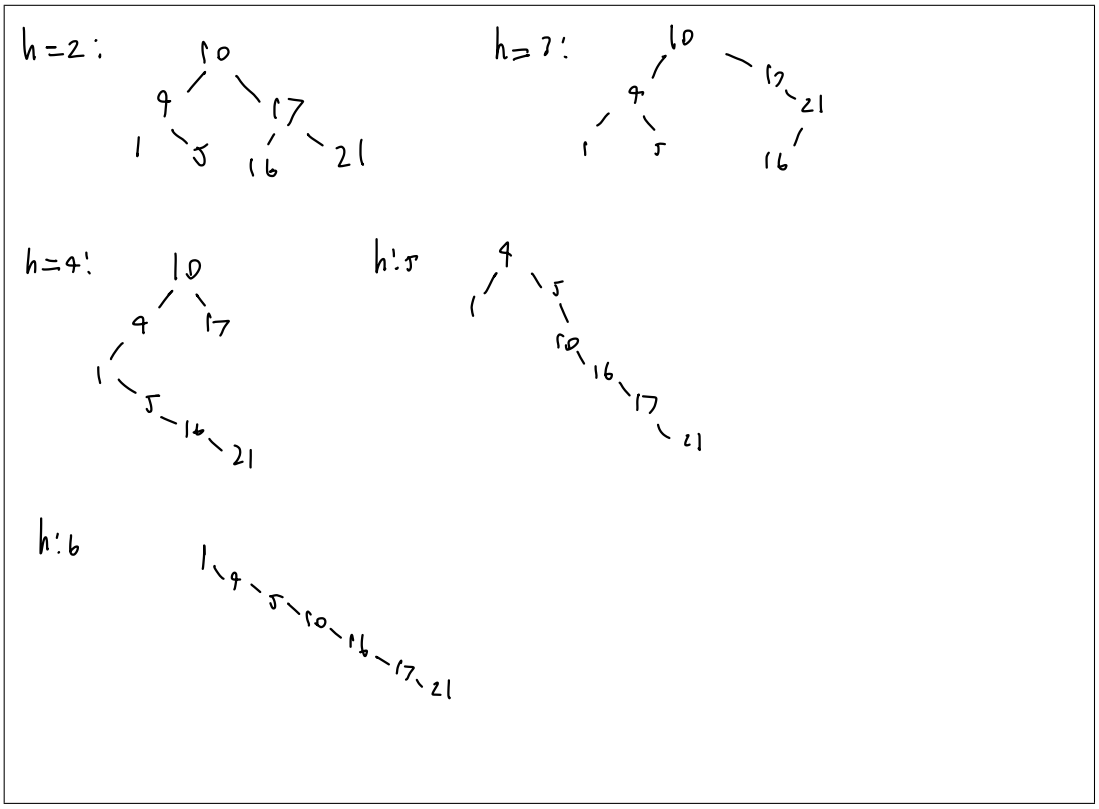
$$A = [15, 13, 9, 5, 12, 8]$$



A = [20, 13, 15, 5, 12, 8, 9]

4. Consider the binary search tree described in chapter 12 of the CLRS textbook.

(a) For the set $\{1, 4, 5, 10, 16, 17, 21\}$ of keys, draw (different, proper) binary search trees with heights 2, 3, 4, 5, and 6.



(b) You are searching for the number 363 in a binary search tree containing numbers between 1 and 1000. Which of the following sequences cannot be the sequence of nodes examined?

i. $2, 252, 401, 398, 330, 344, 397, 363$.
ii. $924, 220, 911, 244, 898, 258, 362, 363$.
iii. $925, 202, 911, 240, 912, 245, 363$.
iv. $2, 399, 387, 219, 266, 382, 381, 278, 363$.
v. $935, 278, 347, 621, 299, 392, 358, 363$.

i) ✓
ii) ✓
iii) ✗
iv) ✓
v) ✗

5. Consider the binary search tree described in chapter 12 of the CLRS textbook.

(a) Show that if a node in a binary search tree has two children, then its successor has no left child.

-successor node is found on the right child then to the left

- if the node is a successor then it will have no child

- if there is a left child, the node is not a successor

- must keep going left

(b) Show that if a node in a binary search tree has two children, then its predecessor has no right child.

-right child nodes will be node that have values higher than it so no predecessor will be found

- predecessor only found on the left side of the tree